

Remote Control of Webcam Through Internet

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Necessity is the mother of invention, and that's how I could find a smart solution for controlling my webcam from a remote location where distance is no limit. This article describes the technique to control the webcam connected to a Raspberry Pi's general-purpose input/output (GPIOs) pins from a remote machine's web browser.

Once I am 800km away from my present location, I may not be able to come back easily for making minor adjustments to the webcam. The system has to run continuously with a webcam connected to it and reliability is of utmost importance. Linux and PHP are reliable tools to achieve my goal. For more than ten years now, I have been using various clones of Linux as my preferred operating system. PHP is a good tool for networking and so I chose my favorite PHP and the smartest solution came in no time.

GPIOs can be manipulated at the raw shell commands level, and PHP can execute shell command almost silently without any ripple on the outside.

The system detail

The Raspberry Pi (Raspi) board streams a live image through a port forwarding it to a remote IP address. The webcam sits on the shaft of a stepper motor, which is being run by the same Raspberry Pi. The prototype is shown in Fig. 1.

The webcam can turn clockwise or anticlockwise, depending upon the remote instructions issued by me. It can even be made to move continuously (150 degrees) from one side to the other and back. This can be done while I am in Howrah and the Raspi webcam

is sitting on my attic in Vindhyanagar, which is about 800km away.

The PHP codes in Raspberry Pi's machine control the webcam smoothly. If there is a sudden power failure, the Raspi will go down but once power comes back, everything will be restored. The camstepper.php program runs at crontab level and restarts at every booting. You can open and edit the camstepper.php using nano editor by typing the following command on the terminal:

```
$ sudo nano camstepper.php
$ sudo su
#crontab -e // this will open the crontab
for su.
```

Add the following line at the bottom; it will start the task at every reboot:

```
#reboot php -f /var/www/camstepper.php
& //
```



Fig. 1: Author's prototype

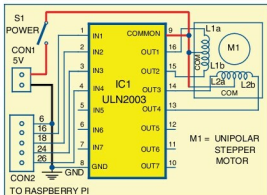


Fig. 2: Circuit diagram and stepper motor connections



Remote control of GPIOs

Ensure that you have apache2 with php5 installed in your Linux system. If not, you can install it as given below:

```
$sudo apt-get install apache2
$sudo apt-get install php5 php5-imap
```

Drop class.phpmailer.php in your document root directory as this file is used for sending email by PHP. The file is available for download from the link: <https://github.com/PHPMailer/PHPMailer/blob/master/class.phpmailer.php>. The imap file is required if you want to install all the features of the webcam. Also, you have to install WiringPi in your system.

Now, let us see the beauty and power of PHP to manipulate the GPIOs of a Raspi board. For controlling the

GPIOs—putting them off or on—we will be using the WiringPi shell commands (gpio) in PHP. For installing the WiringPi, the link is: <https://projects.drogon.net/raspberry-pi/wiringpi/download-and-install/>

Once the WiringPi is installed, you can control the GPIOs by simple command line controls, which we will do in PHP using shell_exec command—so simple and it does so flawlessly in PHP.

The remote machine is the client machine. It can be an Android phone or any computer, but it should be connected to the same network which has the Raspi connected to a webcam. The remote machine should have a browser to open the page to access Raspi's GPIO. The browser should be capable of displaying frames.